

CHAPTER 3

EIGENVALUES AND EIGENVECTORS OF MATRICES

3.1 INTRODUCTION

Eigenvalues play a very important role in obtaining deep understanding of many physical system. For example, the stability of aircraft in flight and the modes of vibration of a bridge are governed by eigenvalues. They are also important in the study of the growth of certain types of population. Suppose X denotes the number of females of a certain age in a given population. If the population is in the state of age stability, the vector X satisfies an equation of the form

$$AX = \lambda X$$

where A is a square matrix and λ is a positive real number which indicates whether the population increases or not. If $\lambda < 1$ the population decreases, If $\lambda > 1$ it increases, and if $\lambda = 1$ it remains the same.

In fact, for a given square matrix A there are only a limited number of values of λ which lead to a non-trivial solution. These are known as the **eigenvalues** of the matrix and the corresponding solution vectors are known as the **eigenvectors** of A .

3.2 EIGENVALUES AND EIGENVECTORS

Let A be a square matrix of order n . A number λ is called an eigenvalue or characteristic value of A if an $n \times 1$ non zero column vector X can be found such that

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3.2 EIGENVALUES AND EIGENVECTORS

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$$AX = \lambda X. \quad (3.1)$$

The above equation may also be written as

$$(A - \lambda I)X = 0$$

If $X^T = (x_1, x_2, \dots, x_n)$, then we have a system of n equations in n unknowns:

$$\begin{aligned} (a_{11} - \lambda)x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n &= 0, \\ a_{21}x_1 + (a_{22} - \lambda)x_2 + \dots + a_{2n}x_n &= 0, \\ \dots & \\ \dots & \\ a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \dots + (a_{nn} - \lambda)x_n &= 0. \end{aligned}$$

It is well-known that a non-trivial solution exists if and only if the determinant of the system vanishes i.e.,

$$\begin{vmatrix} a_{11} - \lambda & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} - \lambda & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} - \lambda \end{vmatrix} = 0 \quad (3.2)$$

This gives an equation of degree n in λ :

$$\lambda^n - (a_{11} + a_{22} + \dots + a_{nn})\lambda^{n-1} + \dots + (-1)^n k = 0, \quad (3.3)$$

where k is yet to be found. It is clear that if we put $\lambda = 0$ in the left hand side of (3.2), we are left with $\det A$. Hence k in (3.3) must be equal to $\det A$. The equation

$$\lambda^n - (a_{11} + a_{22} + \dots + a_{nn})\lambda^{n-1} + \dots + (-1)^n \det A = 0, \quad (3.4)$$

is called the characteristic equation of the matrix A and any root of this equation will be an eigenvalue of A . Thus a matrix of order n has n eigenvalues (although all of them may not be distinct).

EXAMPLE 3.1: Find eigenvalues and corresponding eigenvectors of the matrix

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 3 \end{bmatrix}$$

$$\begin{aligned} \text{SOLUTION: } |A - \lambda I| &= \begin{vmatrix} 2-\lambda & 3 \\ 4 & 3-\lambda \end{vmatrix} = (2 - \lambda)(3 - \lambda) - 12 \\ &= \lambda^2 - 5\lambda - 6. \end{aligned}$$

Thus $\lambda^2 - 5\lambda - 6 = 0$ is the characteristic equation of A and $\lambda = 6, -1$ are the two eigenvalues. To find eigenvectors we have to solve

$$\begin{aligned} (2 - \lambda)x_1 + 3x_2 &= 0, \\ 4x_1 + (3 - \lambda)x_2 &= 0. \end{aligned}$$

First if $\lambda = 6$, then the system becomes

$$\begin{aligned} 4x_1 - 3x_2 &= 0, \\ 4x_1 - 3x_2 &= 0. \end{aligned}$$

So x_2 may be chosen arbitrarily. Let $x_2 = 1$, then $x_1 = 3/4$. So

$$x_1 = \begin{bmatrix} 3/4 \\ 1 \end{bmatrix}$$

is an eigenvector corresponding to $\lambda_1 = 6$. Similarly we find that

$$x_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

is an eigenvector corresponding to $\lambda_2 = -1$.

EXAMPLE 3.2: Find the eigenvalues and the corresponding eigenvectors of

$$A = \begin{bmatrix} -2 & 6 & -24 \\ 0 & -3 & 10 \\ 1 & -4 & 13 \end{bmatrix}$$

$$\text{SOLUTION: } |A - \lambda I| = \begin{vmatrix} -2-\lambda & 6 & -24 \\ 0 & -3-\lambda & 10 \\ 1 & -4 & 13-\lambda \end{vmatrix} = -\lambda^3 + 8\lambda^2 - 5\lambda - 14.$$

The characteristic equation is

$$\lambda^3 - 8\lambda^2 + 5\lambda + 14 = 0,$$

with roots $\lambda = -1, 2, 7$. Hence A has eigenvalues $-1, 2$ and 7 . To find the eigenvectors we proceed as follows. Corresponding to -1 we have to solve

$$AX = -X$$

Let $X = (x_1, x_2, x_3)$. Thus we have to solve the system of equations

$$\begin{aligned} -2x_1 + 6x_2 - 24x_3 &= -x_1, \\ -3x_2 + 10x_3 &= -x_2, \\ x_1 - 4x_2 + 13x_3 &= -x_3, \\ \text{or } x_1 - 6x_2 + 24x_3 &= 0, \\ 2x_2 - 10x_3 &= 0, \\ -x_1 - 4x_2 + 14x_3 &= 0. \end{aligned}$$

Since the equations are not linearly independent, we can choose x_3 arbitrarily. Let $x_3 = 1$, then $x_2 = 5$, $x_1 = 6$. Thus $\begin{bmatrix} 6 \\ 5 \\ 1 \end{bmatrix}$ is an eigenvector corresponding to the eigenvalue $\lambda_1 = -1$. Similarly we find that $\begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$ respectively are the eigenvectors corresponding to the eigenvalues $\lambda_2 = 2$ and $\lambda_3 = 7$.

We now give some simple results about eigenvalues of a matrix. Suppose $\lambda_1, \lambda_2, \dots, \lambda_n$ are the (not necessarily distinct) eigenvalues of a square matrix A .

$$\begin{aligned} \text{THEOREM 3.1: } \lambda_1 + \lambda_2 + \dots + \lambda_n &= a_{11} + a_{22} + \dots + a_{nn} \\ \lambda_1 \lambda_2 \dots \lambda_n &= \det A \end{aligned}$$

This result follows from equation (3.4). The number $\lambda_1 + \lambda_2 + \dots + \lambda_n$ is called the trace of A and is denoted by $\text{tr}(A)$.

THEOREM 3.2: If X is an eigenvector of A corresponding to an eigenvalue λ then so is cX where c is any non zero constant.

PROOF:

$$\begin{aligned} AX &= \lambda X \Rightarrow cAX = c\lambda X \\ \text{or } A(cX) &= \lambda(cX). \end{aligned}$$

DEFINITION 3.1: Two column vectors X_1 and X_2 are said to be orthogonal if $\bar{X}_1 X_2 = 0$.

Here $\bar{X}_1 = (X_1^T)^*$, for example if $X_1 = \begin{bmatrix} 1 \\ 2-i \\ 3 \end{bmatrix}$, then $\bar{X}_1 = (-i, 2+i, 3)$.

DEFINITION 3.2: A matrix is said to be hermitian if $\bar{A} = A$.

EXAMPLE 3.3: If $A = \begin{bmatrix} 1 & 1-i \\ 1+i & 3 \end{bmatrix}$, then $A^T = \begin{bmatrix} 1 & 1+i \\ 1-i & 3 \end{bmatrix}$

and $(A^T)^* = \bar{A} = \begin{bmatrix} 1 & 1-i \\ 1+i & 3 \end{bmatrix} = A$.

Thus A is a hermitian matrix.

THEOREM 3.3: A hermitian matrix has only real eigenvalues.

PROOF: Let A be hermitian and λ be an eigenvalue and X the corresponding eigenvector i.e.

$$AX = \lambda X.$$

By definition X is not the zero vector. Taking the transpose of both sides we have

$$(AX)^T = \lambda X^T,$$

$$\text{or } X^T A^T = \lambda X^T.$$

Take complex conjugates

$$\bar{X} \bar{A} = \bar{\lambda} \bar{X},$$

$$\text{or } \bar{X} A = \bar{\lambda} \bar{X},$$

because A is hermitian. Postmultiply with X

$$\bar{X} A X = \bar{\lambda} \bar{X} X,$$

$$\text{or } \bar{X} \lambda X = \bar{\lambda} \bar{X} X,$$

$$\text{or } (\lambda - \bar{\lambda}) \bar{X} X = 0.$$

But $\bar{X} X \neq 0$, since X is assumed to be a non zero vector. Hence $\lambda = \bar{\lambda}$ which shows that λ is real.

THEOREM 3.4: Eigenvectors corresponding to distinct eigenvalues of a hermitian matrix are orthogonal.

PROOF: Let $AX_1 = \lambda_1 X_1$,
and $AX_2 = \lambda_2 X_2$,

where $\lambda_1 \neq \lambda_2$, and A is a hermitian matrix. Proceeding as above we have

$$\bar{X}_1 A X_2 = \lambda_1 \bar{X}_1 X_2$$

where we used the fact that λ_1 is real. The above equation implies

$$(\lambda_2 - \lambda_1) \bar{X}_1 X_2 = 0.$$

Since $\lambda_1 \neq \lambda_2$, we have

$$\bar{X}_1 X_2 = 0,$$

which proves the theorem.

An important corollary of the above theorem is as follows.

COROLLARY: A symmetric real matrix has only real eigenvalues and the eigenvectors corresponding to distinct eigenvalues are orthogonal.

It is easily shown [8], (page 84) that if $X_1, X_2, \dots, X_n$ are eigenvectors of A corresponding to distinct eigenvalues then the set $\{X_1, \dots, X_n\}$ is linearly independent. Hence every $n \times 1$ column vector X can be expressed as a linear combination of these vectors. However if A is hermitian, it can be shown that [1] even if the eigenvalues are not distinct an orthogonal set of linearly independent eigenvectors $\{X_r\}_{r=1}^n$ can always be found.

EXAMPLE 3.4: The unit matrix of order 3 has an eigenvalue repeated three times. However

$$X_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, X_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, X_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},$$

are three linearly independent orthonormal vectors such that

$$IX_i = X_i, \quad i = 1, 2, 3.$$

EXAMPLE 3.5: The matrix

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

has the eigenvalue unity repeated three times. However there is only one linearly independent eigenvector

$$X = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

3.3 GERSHGORIN'S THEOREM

Now we consider some estimates for the absolute value $|\lambda|$ where λ is any eigenvalue of a square matrix $A = \{a_{ij}\}$. If $X = (x_1, x_2, \dots, x_n)^T$ is an eigenvector corresponding to λ , then we have

$$\lambda x_i = \sum_{r=1}^n a_{ir} x_r, \quad i = 1, 2, \dots, n. \quad (3.5)$$

If $|x_N| = \max_r |x_r|$, then we have

$$\begin{aligned} |\lambda x_N| &= \left| \sum_{r=1}^n a_{Nr} x_r \right| \leq \sum_{r=1}^n |a_{Nr}| |x_r| \\ &\leq \sum_{r=1}^n |a_{Nr}| |x_N| \\ &= |x_N| \sum_{r=1}^n |a_{Nr}| \end{aligned}$$

Hence
$$|\lambda| \leq \sum_{r=1}^n |a_{Nr}|.$$

Since N is, in general, not known we replace the above estimate by a weaker one

$$|\lambda| \leq \max_i \sum_{r=1}^n |a_{ir}| \quad (3.6)$$

Also the transposed matrix has the same set of eigenvalues, hence

$$|\lambda| \leq \max_i \sum_{r=1}^n |a_{ri}|. \quad (3.7)$$

From (3.5) we have

$$(\lambda - a_{NN})x_N = \sum_{r \neq N}^n a_{Nr} x_r.$$

$$\begin{aligned} \text{Thus } |\lambda - a_{NN}| |x_N| &\leq \sum_{r \neq N}^n |a_{Nr}| |x_r| \\ &\leq |x_N| \sum_{r \neq N}^n |a_{Nr}| \end{aligned}$$

$$\text{or } |\lambda - a_{NN}| \leq \sum_{r \neq N}^n |a_{Nr}|.$$

As before we replace the above result by a weaker one: " λ " lies in one of the discs with centre at a_{ii} and radius

$$\sum_{\substack{r=1 \\ r \neq i}}^n |a_{ir}|, \quad i = 1, 2, \dots, n.$$

$$|\lambda - a_{ii}| \leq \sum_{r \neq i}^n |a_{ir}|, \quad i = 1, 2, \dots, n. \quad (3.8)$$

and from the transposed matrix, we have, " λ " lies in one of the discs,

$$|\lambda - a_{ii}| \leq \sum_{r \neq i}^n |a_{ri}|, \quad i = 1, 2, \dots, n. \quad (3.9)$$

All the above estimates hold simultaneously, therefore the best of them is preferred. The above results are known as Gershgorin's Theorem:

THEOREM 3.5: If λ is an eigenvalue of $A = \{a_{ij}\}$ then it satisfies the estimates (3.6) - (3.9).

EXAMPLE 3.6: Consider

$$A = \begin{bmatrix} -2 & 6 & -24 \\ 0 & -3 & 10 \\ 1 & -4 & 13 \end{bmatrix}$$

Here

$$\sum_{r=1}^3 |a_{1r}| = 2 + 6 + 24 = 32$$

$$\sum_{r=1}^3 |a_{2r}| = 0 + 3 + 10 = 13$$

$$\sum_{r=1}^3 |a_{3r}| = 1 + 4 + 13 = 18$$

Therefore (3.6) gives

$$|\lambda| \leq 32. \quad (3.10)$$

Similarly (3.7) gives

$$|\lambda| \leq 47 \quad (3.11)$$

From (3.8) we get

$$|\lambda + 2| \leq 30 \quad \text{or} \quad |\lambda + 3| \leq 10 \quad \text{or} \quad |\lambda - 13| \leq 5. \quad (3.12)$$

Finally (3.9) gives

$$|\lambda + 2| \leq 1 \quad \text{or} \quad |\lambda + 3| \leq 10 \quad \text{or} \quad |\lambda - 13| \leq 34 \quad (3.13)$$

In (3.12) the disc $|\lambda + 2| \leq 30$ contains the other two and in (3.13) $|\lambda - 13| \leq 34$ does the same. Thus

$$|\lambda + 2| \leq 30 \quad \text{is the best estimate.}$$

EXAMPLE 3.7: Consider

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 3 & -2 \\ -5 & 3 & 8 \end{bmatrix}$$

Here we have

$$|\lambda| \leq 16 \quad \text{and} \quad |\lambda| \leq 12.$$

Also $|\lambda - 1| \leq 4$ or $|\lambda - 3| \leq 4$ or $|\lambda - 8| \leq 8$.
 and $|\lambda - 1| \leq 7$ or $|\lambda - 3| \leq 5$ or $|\lambda - 8| \leq 4$.

Thus all eigenvalues lie in the union of $|\lambda - 1| \leq 7$ and $|\lambda - 8| \leq 4$.
 The matrix has eigenvalues 3, 4 and 5.

3.4 THE POWER METHOD

The power method is an iterative technique to find the dominant eigenvalue of a given matrix A . Assume that a matrix has eigenvalues which are so arranged that $|\lambda_1| > |\lambda_2| \geq \dots \geq |\lambda_n|$ and $u_1, u_2, \dots, u_n$ are n linearly independent eigenvectors. If u is any arbitrary $n \times 1$ vector then it can be expressed as a linear combination of the eigenvectors and we write

$$\begin{aligned} u &= a_1 u_1 + a_2 u_2 + \dots + a_n u_n \\ \text{Then } Au &= a_1 A u_1 + a_2 A u_2 + \dots + a_n A u_n \\ &= a_1 \lambda_1 u_1 + a_2 \lambda_2 u_2 + \dots + a_n \lambda_n u_n \end{aligned}$$

It is clear

$$\begin{aligned} A^k u &= a_1 (\lambda_1^k) u_1 + a_2 (\lambda_2^k) u_2 + \dots + a_n (\lambda_n^k) u_n \\ &= \lambda_1^k \left[a_1 u_1 + a_2 \left(\frac{\lambda_2}{\lambda_1} \right)^k u_2 + \dots + a_n \left(\frac{\lambda_n}{\lambda_1} \right)^k u_n \right] \end{aligned} \quad (3.14)$$

Since $\left| \frac{\lambda_r}{\lambda_1} \right| < 1, \quad r = 2, 3, \dots, n,$

the vector in brackets will tend to $a_1 u_1$ as k is taken large. If $(V)_i$ denotes the i -th component of the column vector V , we have

$$\frac{(A^{k+1} u)_i}{(A^k u)_i} = \lambda_1 \left[\frac{[a_1 u_1 + a_2 \left(\frac{\lambda_2}{\lambda_1} \right)^{k+1} u_2 + \dots + a_n \left(\frac{\lambda_n}{\lambda_1} \right)^{k+1} u_n]_i}{[a_1 u_1 + a_2 \left(\frac{\lambda_2}{\lambda_1} \right)^k u_2 + \dots + a_n \left(\frac{\lambda_n}{\lambda_1} \right)^k u_n]_i} \right]$$

Hence
$$\lim_{k \rightarrow \infty} \frac{(A^{k+1}u)_1}{(A^k u)_1} = \lambda_1 \frac{a_1(u_1)_1}{a_1(u_1)_1} = \lambda_1 \quad (3.15)$$

From (3.15) we have for large k

$$A^{k+1}u \approx \lambda_1 A^k u$$

Thus
$$A(A^k u) \approx \lambda_1 (A^k u)$$

Therefore $A^k u$ is an approximate eigenvector corresponding to the eigenvalue λ_1 . The method based on the above analysis, for finding the dominant (in absolute value) eigenvalue of a matrix, is called the power method. We proceed as follows:

- Choose a suitable arbitrary column vector u . Usually u is chosen as $(1, 1, \dots, 1)^T$.
- Form $v = Au$.
- Multiply v by a number k_1 so as to make the largest element equal to 1. If one is working with an automatic machine, it is easier to multiply, at every step, with

$$k_1 = \left(|v_1|^2 + |v_2|^2 + \dots + |v_n|^2 \right)^{-1/2}.$$

We do this because if v is an eigenvector then so is kv . This is advisable to stop the numbers from getting too large.

- Iterate $v_{i+1} = A k_1 v_i \quad i = 1, 2, \dots$

EXAMPLE 3.8: Find the dominant eigenvalue of

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

SOLUTION: Let

$$u = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \text{ then } v = Au = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad u_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Denote by u_1 the normalized column vector $k_1 v_1$ i.e. the vector whose largest element in absolute value has been made equal to 1.

$$v_2 = A u_1 = \begin{bmatrix} 2 \\ -2 \\ 2 \end{bmatrix}$$

$$u_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$v_3 = A u_2 = \begin{bmatrix} 3 \\ -4 \\ 3 \end{bmatrix}$$

$$u_3 = \begin{bmatrix} -0.75 \\ 1 \\ -0.75 \end{bmatrix}$$

$$v_4 = A u_3 = \begin{bmatrix} -2.5 \\ 3.5 \\ -2.5 \end{bmatrix}$$

$$u_4 = \begin{bmatrix} -0.75 \\ 1 \\ -0.75 \end{bmatrix}$$

$$v_5 = A u_4 = \begin{bmatrix} -2.428 \\ 3.428 \\ -2.428 \end{bmatrix}$$

$$u_5 = \begin{bmatrix} -0.7083 \\ 1 \\ -0.7083 \end{bmatrix}$$

$$v_6 = A u_5 = \begin{bmatrix} -2.4166 \\ 3.4166 \\ -2.4166 \end{bmatrix}$$

$$u_6 = \begin{bmatrix} -0.7073 \\ 1 \\ -0.7073 \end{bmatrix}$$

$$v_7 = A u_6 = \begin{bmatrix} -2.4146 \\ 3.4146 \\ -2.4146 \end{bmatrix}$$

$$u_7 = \begin{bmatrix} -0.70714 \\ 1 \\ -0.70714 \end{bmatrix}$$

$$v_8 = A u_7 = \begin{bmatrix} -2.41428 \\ 3.41428 \\ -2.41428 \end{bmatrix}$$

$$u_8 = \begin{bmatrix} -0.707112 \\ 1 \\ -0.707112 \end{bmatrix}$$

$$v_9 = A u_8 = \begin{bmatrix} -2.414224 \\ 3.414224 \\ -2.414224 \end{bmatrix}$$

$$u_9 = \begin{bmatrix} -0.70711 \\ 1 \\ -0.70711 \end{bmatrix}$$

Hence to 5 decimal places

$$A u_8 = 3.414224 \begin{bmatrix} -0.70711 \\ 1 \\ -0.70711 \end{bmatrix} = 3.414224 u_9$$

Therefore the dominant eigenvalue is 3.414224 and the corresponding eigenvector is $(-0.70711, 1, -0.70711)^T$.

The exact value is $2 + \sqrt{2} = 3.414214$.

The smallest and the largest eigenvalues:

We can modify the power method to find the smallest and the

largest eigenvalues.

Let λ be an eigenvalue of a matrix A and X be the corresponding eigenvector. Then

$$AX = \lambda X$$

Subtracting αX from both sides of the above equation we get

$$\begin{aligned} AX - \alpha X &= \lambda X - \alpha X \\ (A - \alpha I)X &= (\lambda - \alpha)X \end{aligned}$$

Thus $\lambda - \alpha$ is an eigenvalue of the matrix $A - \alpha I$ and X is the associated vector. In other words, the effect of subtracting α from each element of the leading diagonal of a matrix has the effect of subtracting α from each eigenvalue but leaves the eigenvectors unchanged. Therefore, we can make, by a proper choice of α , the largest or the smallest eigenvalue the dominant one in absolute value.

Suppose that α is taken equal to or greater than the dominant eigenvalue of A . Then, if the dominant eigenvalue of $A - \alpha I$ is found, adding α to it will give the numerically smallest eigenvalue of A . But the corresponding eigenvector will be the same as that found for the dominant eigenvalue of $A - \alpha I$.

EXAMPLE 3.9: Find the smallest eigenvalue of

$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

SOLUTION: It is clear that all eigenvalues satisfy $|\lambda - 2| \leq 2$. Thus they must lie in the interval $[0, 4]$. Hence the largest eigenvalue is also the dominant (in absolute value). To find the smallest eigenvalue we consider the matrix

$$B = A - 4I = \begin{bmatrix} -2 & -1 & 0 \\ -1 & -2 & -1 \\ 0 & -1 & -2 \end{bmatrix}$$

Apply the power method to B .

$$\text{Let } u = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \text{ then } v = Au = \begin{bmatrix} -3 \\ -4 \\ -3 \end{bmatrix} = -4 \begin{bmatrix} 0.75 \\ 1 \\ 0.75 \end{bmatrix} = -4u_1$$

$$v_2 = Au_1 = \begin{bmatrix} -2.5 \\ -3.5 \\ -2.5 \end{bmatrix} = -3.5 \begin{bmatrix} 0.7143 \\ 1 \\ 0.7143 \end{bmatrix} = -3.5 u_2$$

$$v_3 = Au_2 = \begin{bmatrix} -2.4286 \\ -3.4286 \\ -2.4286 \end{bmatrix} = -3.4286 \begin{bmatrix} 0.70833 \\ 1 \\ 0.70833 \end{bmatrix} = -3.4286 u_3$$

$$v_4 = Au_3 = \begin{bmatrix} -2.41667 \\ -3.41667 \\ -2.41667 \end{bmatrix} = -3.41667 \begin{bmatrix} 0.70732 \\ 1 \\ 0.70732 \end{bmatrix} = -3.41667 u_4$$

$$v_5 = Au_4 = \begin{bmatrix} -2.41464 \\ -3.41464 \\ -2.41464 \end{bmatrix} = -3.41464 \begin{bmatrix} 0.7071434 \\ 1 \\ 0.7071434 \end{bmatrix} = -3.41464 u_5$$

$$v_6 = Au_5 = \begin{bmatrix} -2.4142868 \\ -3.4142868 \\ -2.4142868 \end{bmatrix} = -3.4142868 \begin{bmatrix} 0.7071131 \\ 1 \\ 0.7071131 \end{bmatrix} = -3.4142868 u_6$$

$$v_7 = Au_6 = \begin{bmatrix} -2.414226127 \\ -3.414226127 \\ -2.414226127 \end{bmatrix} = -3.414226 \begin{bmatrix} 0.707108 \\ 1 \\ 0.707108 \end{bmatrix} = -3.414226 u_7$$

Thus an approximate eigenvalue for B is -3.414226. The corresponding eigenvalue for A is $-3.414226 + 4 = 0.585774$.

The exact value is $2 - \sqrt{2} = 0.58578644$. The approximate corresponding eigenvector for this eigenvalue is

$$\begin{bmatrix} 0.707108 \\ 1 \\ 0.707108 \end{bmatrix}$$

The question arises that whether the smallest eigenvalue can be found without finding the largest one. Suppose

$$AX = \lambda X$$

then $X = A^{-1}\lambda X = \lambda A^{-1}X$ (As λ is scalar)

or $A^{-1}X = \lambda^{-1}X.$

This equation shows that $1/\lambda$ is an eigenvalue of A^{-1} . If the power method is used to find the largest eigenvalue $1/\lambda$ of A^{-1} then the smallest eigenvalue of A is λ . Moreover, if X is the eigenvector of A^{-1} corresponding to eigenvalue $1/\lambda$, then X is also the eigenvector of A corresponding to eigenvalue λ .

EXAMPLE 3.10: Let $A = \begin{bmatrix} 5 & 2 \\ 2 & 8 \end{bmatrix}$, then

$$A^{-1} = \frac{1}{36} \begin{bmatrix} 8 & -2 \\ -2 & 5 \end{bmatrix} = \begin{bmatrix} 0.2222 & -0.0556 \\ -0.0556 & 0.1389 \end{bmatrix}$$

By power method we can find the largest eigenvalue of A^{-1} which is 0.25. The corresponding eigenvector is $(1, -0.5)^T$. The eigenvalue of A is $1/0.25$ or 4 and the corresponding eigenvector is $(1, -0.5)^T$. The eigenvalues of A calculated analytically are 4, 9. Thus both the results agree.

Rayleigh Quotient: If a matrix is Hermitian its eigenvectors $u_1, u_2, \dots, u_n$ are orthogonal. We suppose that we have also made them orthonormal. Let u be an arbitrary column vector such that

$$\bar{u}u_1 \neq 0. \text{ Now}$$

$$u = c_1 u_1 + c_2 u_2 + \dots + c_n u_n$$

$$v_p = A^p u = c_1 (\lambda_1)^p u_1 + c_2 (\lambda_2)^p u_2 + \dots + c_n (\lambda_n)^p u_n$$

and

$$v_{p+1} = c_1 (\lambda_1)^{p+1} u_1 + \dots + c_n (\lambda_n)^{p+1} u_n$$

Hence using the orthonormality property of the eigenvectors i.e.

$$\bar{u}_i u_j = \delta_{ij}, \text{ we find}$$

$$\gamma_p = \frac{\bar{v}_{p+1} v_p}{\bar{v}_p v_p} = \lambda_1 \frac{|c_1|^2 + |c_2|^2 \left(\frac{\lambda_2}{\lambda_1}\right)^{2p+1} + \dots + |c_n|^2 \left(\frac{\lambda_n}{\lambda_1}\right)^{2p+1}}{|c_1|^2 + |c_2|^2 \left(\frac{\lambda_2}{\lambda_1}\right)^{2p} + \dots + |c_n|^2 \left(\frac{\lambda_n}{\lambda_1}\right)^{2p}}$$

Thus we see that the convergence of the sequence $\{\gamma_p\}_{p=1}^{\infty}$ to λ_1 is much faster as compared with the sequence given by (3.15). This is called the sequence of Rayleigh quotients. For example in the example 3.9,

$$\gamma_1 = -3.3333,$$

$$\gamma_2 = -3.4117647, \dots, \gamma_6 = -3.414213473$$

and we see that γ_6 is much closer to the exact value -3.414213562 as compared with the value -3.414226 obtained by the earlier method.

3.5 DEFLATION

Suppose we have found an eigenvalue λ_1 and the corresponding eigenvector X_1 of an $n \times n$ matrix A . Deflation means finding a matrix of order $(n-1) \times (n-1)$ whose eigenvalues coincide with the rest of the eigenvalues of A . Suppose $X_1 = (x_1, x_2, \dots, x_n)^T$. We assume that $x_1 \neq 0$ and multiply X_1 by $1/x_1$ to get an eigenvector X_1^* whose top element is unity i.e.

$$X_1^* = (1, x_2^*, \dots, x_n^*)^T.$$

If the i -th row of A is denoted by R_i , we see that

$$AX_1^* = \begin{bmatrix} R_1 X_1^* \\ R_2 X_1^* \\ R_3 X_1^* \\ \vdots \\ R_n X_1^* \end{bmatrix}.$$

Since $AX_1^* = \lambda_1 X_1^* = (\lambda_1, \lambda_1 x_2^*, \dots, \lambda_1 x_n^*)^T$, it is clear that

$$R_1 X_1^* = \lambda_1 \quad (3.16a)$$

If rest of the eigenvectors have also been normalized in the same manner i.e. their top element is 1, then

$$R_i X_i^* = \lambda_i \quad (3.16b)$$

If, however X_1^* is such that its top element is zero, then

$$R_1 X_1^* = 0. \quad (3.16c)$$

Now we form the matrix B defined as

$$B = A - X_1^* R_1.$$

Note that $X_1^* R_1$ is a an $n \times n$ matrix whose first row is R_1 , hence B has only zeros in its first row. We show that with the exception of λ_1 , matrices A and B have all eigenvalues common.

THEOREM 3.6: If A has eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ then B has eigenvalues 0, $\lambda_2, \dots, \lambda_n$.

PROOF: Suppose $AX_1^* = \lambda_1 X_1^*$ ($i = 1, 2, \dots, n$)

where X_1^* has its top element equal to unity and each of $X_2^*, X_3^*, \dots, X_n^*$ has its top element either 1 or 0. Now

$$\begin{aligned} BX_1^* &= (A - X_1^* R_1) X_1^* \\ &= AX_1^* - X_1^* (R_1 X_1^*) \\ &= \lambda_1 X_1^* - X_1^* \lambda_1 \\ &= 0. \end{aligned}$$

Hence 0 is an eigenvalue of B with eigenvector X_1^* . If X_1^* has its top element unity then

$$\begin{aligned} B(X_1^* - X_1^*) &= (A - X_1^* R_1)(X_1^* - X_1^*) \\ &= AX_1^* - AX_1^* - X_1^* (R_1 X_1^*) + X_1^* (R_1 X_1^*) \\ &= \lambda_1 X_1^* - \lambda_1 X_1^* - \lambda_1 X_1^* + \lambda_1 X_1^* \\ &= \lambda_1 (X_1^* - X_1^*). \end{aligned} \quad (3.17)$$

In the above, we have made use of (3.16a) and (3.16b). Now it is clear from (3.17) that λ_1 is an eigenvalue of B and $X_1^* - X_1^*$ is the

corresponding eigenvector. Finally if X_1^* has its top element equal to zero, then

$$\begin{aligned} B X_j^* &= A X_j^* - X_1^* R_1 X_j^* \\ &= A X_j^* \end{aligned}$$

Since $R_1 X_j^* = 0$ by (3.16c). Thus again we have

$$B X_j^* = \lambda_j X_j^* \quad (3.18)$$

This completes the proof.

Now define an $(n-1) \times (n-1)$ matrix A_1 obtained by deleting the first row and the first column of B . Let

$$B = \begin{bmatrix} 0 & 0 & 0 & \dots & 0 \\ b_{21} & b_{22} & b_{23} & \dots & b_{2n} \\ b_{31} & b_{32} & b_{33} & \dots & b_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & b_{n3} & \dots & b_{nn} \end{bmatrix}$$

Then

$$A_1 = \begin{bmatrix} b_{22} & b_{23} & b_{24} & \dots & b_{2n} \\ b_{32} & b_{33} & b_{34} & \dots & b_{3n} \\ b_{42} & b_{43} & b_{44} & \dots & b_{4n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b_{n2} & b_{n3} & b_{n4} & \dots & b_{nn} \end{bmatrix}$$

Since

$$|B - \lambda I_n| = -\lambda |A_1 - \lambda I_{n-1}|, \quad (3.19)$$

it is obvious that B and A_1 have all eigenvalues common with the exception of $\lambda = 0$. In (3.19) I_n denotes the unit matrix of order n . A_1 is the so called deflated matrix we were seeking.

EXAMPLE 3.11: Let $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 3 & -2 \\ -5 & 3 & 8 \end{bmatrix}$.

A has an eigenvalue 4 with $(2, 2, 1)^T$ as an eigenvector. Here $\lambda_1 = 4$, $X_1^* = (1, 1, 1/2)^T$ and $R_1 = (1, 2, 2)$.
Now

$$X_1^* R_1 = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 2 & 2 \\ 1/2 & 1 & 1 \end{bmatrix},$$

and $B = A - X_1^* R_1$ comes out as

$$B = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & -4 \\ -\frac{1}{2} & 2 & 7 \end{bmatrix}$$

The matrix A_1 obtained by deleting the first row and the first column of B is found to be

$$A_1 = \begin{bmatrix} 1 & -4 \\ 2 & 7 \end{bmatrix}$$

Now $|A_1 - \lambda I| = \lambda^2 - 8\lambda + 15$
 $= (\lambda - 5)(\lambda - 3).$

Thus A_1 has eigenvalues 5 and 3. These are also the remaining eigenvalues of A .

We can recover eigenvectors of the matrix A from those of deflated matrix A_1 in a simple manner. First we note that if

$(y_1, y_2, \dots, y_{n-1})^T$ is an eigenvector of matrix A_1 then $(0, y_1, y_2, \dots, y_{n-1})^T$ is an eigenvector of the matrix B . Let us denote this vector by Y_1 and suppose the corresponding eigenvalue is λ_1 , i.e.

$$B Y_1 = \lambda_1 Y_1$$

(3.20)

However from (3.17) and (3.18) we have

$$B(X_1^* - X_1^*) = \lambda_1(X_1^* - X_1^*) \quad (3.21)$$

if $R_1 X_1^* = \lambda_1 \neq 0$

and $BX_1^* = \lambda_1 X_1^*$

if $R_1 X_1^* = 0. \quad (3.22)$

If (3.21) holds then Y_1 must be proportional to $X_1^* - X_1^*$ since these are eigenvectors of B corresponding to the same eigenvalue. Thus for some α

$$X_1^* - X_1^* = \alpha Y_1. \quad (3.23)$$

Multiply (3.23) with R_1 , we get

$$R_1 X_1^* - R_1 X_1^* = \alpha R_1 Y_1,$$

or $\lambda_1 - \lambda_1 = \alpha R_1 Y_1.$

This gives $\alpha = (\lambda_1 - \lambda_1)/R_1 Y_1$ and (3.23) gives

$$X_1^* = X_1^* - (\lambda_1 - \lambda_1)Y_1/R_1 Y_1. \quad (3.24)$$

If $R_1 X_1^* = 0$, i.e. (3.22) holds, then

$$Y_1 = \beta X_1^*$$

for some $\beta \neq 0$, since X_1^* and Y_1 happen to be eigenvectors of B corresponding to the same eigenvalue. In this case Y_1 itself is an eigenvector of A . Actually we do not know X_1^* , however $R_1 Y_1^* \neq 0$ implies $R_1 X_1^* \neq 0$ and $R_1 Y_1^* = 0$ implies $R_1 X_1^* = 0$.

EXAMPLE 3.12: Let $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 3 & -2 \\ -5 & 3 & 8 \end{bmatrix}$.

We know that $\lambda_1 = 4$, $X_1^* = (1, 1, 1/2)^T$. We found that the deflated matrix A_1 came out to be

$$A_1 = \begin{bmatrix} 1 & -4 \\ 2 & 7 \end{bmatrix}.$$

A_1 has an eigenvalue 5 with an eigenvector $(1, -1)^T$ and an eigenvalue 3 with an eigenvector $(2, -1)^T$. Let us take $\lambda_2 = 5$ and

$$Y_2 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}.$$

Since $R_1 Y_2 = 0$, we conclude that Y_2 is an eigenvector of A . Now let $\lambda_3 = 3$ and

$$Y_3 = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}.$$

Here $R_1 Y_3 = 2$ and (3.24) gives

$$\begin{aligned} X_3^* &= \begin{bmatrix} 1 \\ 1 \\ 1/2 \end{bmatrix} - \frac{(4-3)}{2} \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}. \end{aligned}$$

Thus we have finally found that the matrix A has eigenvalues 4, 5 and 3 with respective eigenvectors $(1, 1, 1/2)^T$, $(0, 2, -1)^T$ and $(1, 0, 1)^T$.

In the beginning of this section we assumed that the top element of X_1 is different from zero. If it happens to be zero we make the second element from the top unity by multiplying with a suitable constant and carry through with the analysis with minor modifications.

In practice we can find the dominant eigenvalue and the corresponding eigenvector of a given matrix A by the power method, then

find the matrix A_1 by deflation, apply the power method to A_1 to find the next dominant eigenvalue and so on.

In the case of 3×3 matrix we can find the dominant eigenvalue and the corresponding eigenvector of the matrix by the power method, then the smallest eigenvalue and the corresponding vector. The third eigenvalue follows immediately as the sum of the eigenvalues is equal to the trace of the given matrix.

The power method fails if the dominant eigenvalues are a pair of complex conjugate numbers. Also the convergence is poor if ratio of the two dominant eigenvalues is close to unity. In such a case Jacobi's method may be used. For details see Fröberg [4].

EXERCISE 3

- Find the eigenvalues and the corresponding eigenvectors of the following matrices analytically.

$$A = \begin{bmatrix} 1 & 2 & -8 \\ -2 & 2 & -2 \\ 1 & -4 & 10 \end{bmatrix}, B = \begin{bmatrix} 0 & 2 & -10 \\ -3 & 1 & -3 \\ 1 & -5 & 11 \end{bmatrix}, C = \begin{bmatrix} -2 & 2 & 2 \\ 3 & -1 & 3 \\ 1 & 1 & -3 \end{bmatrix}.$$

- Find the dominant eigenvalue and the corresponding eigenvector for each of the following matrices. Give your answers correct to three decimal places.

$$(a) \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 1 & 1 & 1 \end{bmatrix} \quad (b) \begin{bmatrix} -2 & 1 & 0 \\ 1 & -2 & 1 \\ 0 & 1 & -2 \end{bmatrix}.$$

- Find all the eigenvalues of the matrix

$$\begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

by calculating the roots of its characteristic equation. Hence determine the associated eigenvectors and normalize them. Give your answers correct to four decimal places.

- Taking $u = (0,0,1)^T$, use the power method to evaluate the dominant eigenvalue and corresponding eigenvector of the matrix A given in Exercise 3. Give your answer correct to four decimal places and compare them with those obtained previously.

5. Using the estimate of eigenvector obtained in Exercise 4, deflate the matrix A and hence calculate the remaining eigenvalues and associated eigenvectors. Compare your results with those obtained in Exercise 3.
6. Using the estimate of eigenvector obtained in Exercise 4, find the smallest eigenvalue and associated eigenvector of A .
7. Consider the matrix

$$A = \begin{bmatrix} 0 & 2 & -8 \\ -2 & 1 & -2 \\ 1 & -4 & 9 \end{bmatrix}$$

having $\lambda_1 = -1$ as an eigenvalue and $X_1 = (1, 3/2, 1/2)^T$ the corresponding eigenvector. Deflate the matrix and hence calculate the remaining eigenvalues and associated eigenvectors.

CHAPTER 4

INTERPOLATION WITH UNEQUALLY SPACED DATA

4.1 INTRODUCTION

Suppose we are given the values of a function $f(x)$ at certain points $x_0, x_1, \dots, x_n$ and we want to estimate $f(\alpha)$ where α is any point in the interval $[x_0, x_n]$. We assume that the function can be approximated by a polynomial. This is called polynomial interpolation.

There are a number of interpolation formulae, most of which have certain advantages over the others in certain situations, but no one of which is preferable to all others in all respects. In the general case, when the abscissas are not equally spaced, the use of divided differences is convenient.

4.2 LAGRANGE'S FORMULA

We assume that the polynomial

$$P(x) = a_0 + a_1x + \dots + a_nx^n$$

is such that $P(x_r) = y_r$, $r = 0, 1, 2, \dots, n$. Thus the curves representing graphs of the two functions $y = f(x)$ and $y = P(x)$ pass through the same $n+1$ points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$. If the function $f(x)$ has continuous derivatives of all orders, it is reasonable to suppose that $f(x)$ and $P(x)$ will be fairly close at other points of the interval $[x_0, x_n]$. To determine $a_0, a_1, \dots, a_n$ we have to solve the system of equations:

$$\left. \begin{aligned}
 a_0 + a_1 x_0 + a_2 x_0^2 + \dots + a_n x_0^n &= y_0, \\
 a_0 + a_1 x_1 + a_2 x_1^2 + \dots + a_n x_1^n &= y_1, \\
 \dots &\dots \dots \dots \dots \dots \\
 a_0 + a_1 x_n + a_2 x_n^2 + \dots + a_n x_n^n &= y_n.
 \end{aligned} \right\} \quad (4.1)$$

The determinant of the system is

$$\begin{vmatrix}
 1 & x_0 & x_0^2 & \dots & x_0^n \\
 1 & x_1 & x_1^2 & \dots & x_1^n \\
 \dots & \dots & \dots & \dots & \dots \\
 1 & x_n & x_n^2 & \dots & x_n^n
 \end{vmatrix}$$

Since $x_0, x_1, \dots, x_n$ are all distinct, this determinant is not zero and the system (4.1) has a unique solution. However one or more, but not all, of a_n may turn out to be zero. Thus we have shown that a polynomial of degree $\leq n$ exists such that the curve representing $y = P(x)$ passes through the given $n+1$ points. Also this polynomial is unique. Let us define

$$L_k(x) = \frac{(x-x_0)(x-x_1)\dots(x-x_{k-1})(x-x_{k+1})\dots(x-x_n)}{(x_k-x_0)(x_k-x_1)\dots(x_k-x_{k-1})(x_k-x_{k+1})\dots(x_k-x_n)}.$$

We see that

$$L_k(x_r) = \begin{cases} 0, & \text{if } r \neq k, \\ 1, & \text{if } r = k. \end{cases} \quad (4.2)$$

If we write

$$P(x) = L_0(x)y_0 + L_1(x)y_1 + \dots + L_n(x)y_n, \quad (4.3)$$

we see that $P(x)$ is a polynomial of degree $\leq n$ (since $L_0(x), L_1(x), \dots$ are polynomials of degree n and coefficients of x^n, x^{n-1} etc.

$$\left. \begin{aligned}
 a_0 + a_1 x_0 + a_2 x_0^2 + \dots + a_n x_0^n &= y_0, \\
 a_0 + a_1 x_1 + a_2 x_1^2 + \dots + a_n x_1^n &= y_1, \\
 \dots &\dots \dots \dots \dots \dots \\
 a_0 + a_1 x_n + a_2 x_n^2 + \dots + a_n x_n^n &= y_n.
 \end{aligned} \right\} \quad (4.1)$$

The determinant of the system is

$$\begin{vmatrix}
 1 & x_0 & x_0^2 & \dots & x_0^n \\
 1 & x_1 & x_1^2 & \dots & x_1^n \\
 \dots & \dots & \dots & \dots & \dots \\
 1 & x_n & x_n^2 & \dots & x_n^n
 \end{vmatrix}$$

Since $x_0, x_1, \dots, x_n$ are all distinct, this determinant is not zero and the system (4.1) has a unique solution. However one or more, but not all, of a_n may turn out to be zero. Thus we have shown that a polynomial of degree $\leq n$ exists such that the curve representing $y = P(x)$ passes through the given $n+1$ points. Also this polynomial is unique. Let us define

$$L_k(x) = \frac{(x-x_0)(x-x_1)\dots(x-x_{k-1})(x-x_{k+1})\dots(x-x_n)}{(x_k-x_0)(x_k-x_1)\dots(x_k-x_{k-1})(x_k-x_{k+1})\dots(x_k-x_n)}.$$

We see that

$$L_k(x_r) = \begin{cases} 0, & \text{if } r \neq k, \\ 1, & \text{if } r = k. \end{cases} \quad (4.2)$$

If we write

$$P(x) = L_0(x)y_0 + L_1(x)y_1 + \dots + L_n(x)y_n, \quad (4.3)$$

we see that $P(x)$ is a polynomial of degree $\leq n$ (since $L_0(x), L_1(x), \dots$ are polynomials of degree n and coefficients of x^n, x^{n-1} etc.

may add up to zero), also

$$P(x_0) = y_0, P(x_1) = y_1, \dots, P(x_n) = y_n.$$

Thus $P(x)$ is the required polynomial. Equation (4.3) is called the Lagrange's interpolation formula. Note that, although we proved the existence of such a polynomial by a different method, we seldom use the determinant method to actually evaluate the constants $a_0, a_1, \dots, a_n$. Later we shall discuss other methods for finding the interpolation polynomial. Since such a polynomial is unique, it does not matter how we find it. The result is always the same.

EXAMPLE 4.1: Let values of $y = f(x)$ be as given in the following table. Find a polynomial which interpolates the given data

x	y
-1	-8
1	-2
2	4
3	28

SOLUTION: Here $x_0 = -1, x_1 = 1, x_2 = 2, x_3 = 3,$

$$y_0 = -8, y_1 = -2, y_2 = 4, y_3 = 28.$$

Thus

$$L_0(x) = \frac{(x-1)(x-2)(x-3)}{(-1-1)(-1-2)(-1-3)},$$

$$L_1(x) = \frac{(x-(-1))(x-2)(x-3)}{(1-(-1))(1-2)(1-3)} \text{ etc.}$$

Here

$$\begin{aligned} P(x) &= \frac{(x-1)(x-2)(x-3)}{(-2)(-3)(-4)}(-8) + \frac{(x+1)(x-2)(x-3)}{2(-1)(-2)}(-2) \\ &\quad + \frac{(x-(-1))(x-1)(x-3)}{(2-(-1))(2-1)(2-3)}(4) + \frac{(x-(-1))(x-1)(x-2)}{(3-(-1))(3-1)(3-2)}(28) \\ &= (x^3 - 6x^2 + 11x - 6)(1/3) + (x^3 - 4x^2 + x + 6)(-1/2) \\ &\quad + (x^3 - 3x^2 - x + 3)(-4/3) + (x^3 - 2x^2 - x + 2)(7/2) \\ &= 2x^3 - 3x^2 + x - 2. \end{aligned}$$

4.3 THE ERROR OF THE INTERPOLATION POLYNOMIAL

Suppose we want to consider the error $R(x)=f(x)-P(x)$ at a point α in the interval $[x_0, x_n]$. Of course $R=0$ if $\alpha = x_0, x_1, \dots, x_n$. So let α be a point distinct from them. Let us define

$$F(x) = (x - x_0)(x - x_1) \dots (x - x_n)$$

and write

$$g(x) = f(x) - p(x) - KF(x). \quad (4.4)$$

In (4.4) K is a constant to be chosen in a manner to make $g(\alpha)=0$. Note that $g(x_0) = g(x_1) = \dots = g(x_n) = 0$.

Consider $g(x)$ on the interval $[x_0, x_1]$. Now $g(x_0)=g(x_1)=0$, also g is continuous on $[x_0, x_1]$ and differentiable on (x_0, x_1) hence, by Rolle's theorem, there is a point β_1 in (x_0, x_1) such that $g'(\beta_1) = 0$. Applying the same argument to $g(x)$ on intervals $[x_1, x_2], \dots, [x_{n-1}, x_n]$ we find that there exist points $\beta_2, \beta_3, \dots, \beta_{n+1}$ such that the derivative of $g(x)$ vanishes at each of these points. Now we apply Rolle's theorem to $g'(x)$ and find that there exist points $\gamma_1, \gamma_2, \dots, \gamma_n$ such that $g''(x)$ vanishes at each of these points. Proceeding in this manner, we find that there exists a point ξ in (x_0, x_n) such that $g^{(n+1)}(\xi)=0$. Now from (4.4)

$$g^{(n+1)}(\xi) = f^{(n+1)}(\xi) - P^{(n+1)}(\xi) - K(n+1)!$$

But $P^{(n+1)}(\xi) = 0$, because $P(x)$ is a polynomial of degree at most n . Hence

$$K = \frac{f^{(n+1)}(\xi)}{(n+1)!}$$

From (4.4) we get, on putting $x = \alpha$ (remember $g(\alpha) = 0$)

$$f(\alpha) = P(\alpha) + \frac{f^{(n+1)}(\xi)}{(n+1)!} F(\alpha). \quad (4.5)$$

In the beginning we assumed that $\alpha \neq x_0, x_1, \dots, x_n$. However (4.5) is obviously satisfied even if we remove this restriction. Since α is an arbitrary point we can write

$$f(x) = P(x) + \frac{f^{(n+1)}(\xi)}{(n+1)!} F(x), \quad x \in [x_0, x_n].$$

Note that ξ also depends on x . We have shown that the error $f(x) - P(x)$ committed by approximating the function $f(x)$ by the interpolation polynomial is

$$R = \frac{f^{(n+1)}(\xi) F(x)}{(n+1)!}, \quad x_0 < \xi < x_n. \quad (4.6)$$

4.4 DIVIDED DIFFERENCES

Let the values of a function f be given at $n+1$ points $x_0, x_1, \dots, x_n$. We define the first divided difference of $f(x)$ between x_0 and x_1 as

$$f(x_0, x_1) = \frac{f(x_1) - f(x_0)}{x_1 - x_0}. \quad (4.7)$$

Note that $f(x_0, x_1) = f(x_1, x_0)$. We define the second divided difference, $f(x_0, x_1, x_2)$, as

$$f(x_0, x_1, x_2) = \frac{f(x_1, x_2) - f(x_0, x_1)}{x_2 - x_0}.$$

Similarly the n -th divided difference is defined as

$$f(x_0, x_1, \dots, x_n) = \frac{f(x_1, x_2, \dots, x_n) - f(x_0, x_1, \dots, x_{n-1})}{x_n - x_0}. \quad (4.8)$$

Now we shall prove the following theorem:

THEOREM 4.1:

$$f(x_0, x_1, \dots, x_n) = \sum_{r=0}^n \frac{f(x_r)}{(x_r - x_0)(x_r - x_1) \dots (x_r - x_{r-1})(x_r - x_{r+1}) \dots (x_r - x_n)}$$

PROOF: Obviously the result is true for $n = 1$, (see 4.7). Let the result be true for $n = k$. Thus we have

$$f(x_0, x_1, \dots, x_k) = \sum_{r=0}^k \frac{f(x_r)}{(x_r - x_0)(x_r - x_1) \dots (x_r - x_{r-1})(x_r - x_{r+1}) \dots (x_r - x_k)} \quad (4.9)$$

Now by definition

$$\begin{aligned} f(x_0, x_1, \dots, x_{k+1}) &= \frac{f(x_1, \dots, x_{k+1}) - f(x_0, \dots, x_k)}{x_{k+1} - x_0} \\ &= \frac{1}{(x_{k+1} - x_0)} \left[\sum_{r=1}^{k+1} \frac{f(x_r)}{(x_r - x_1) \dots (x_r - x_{k+1})} - \sum_{r=0}^k \frac{f(x_r)}{(x_r - x_0) \dots (x_r - x_k)} \right] \end{aligned}$$

where we have used Eq. (4.9),

$$\begin{aligned} &= \frac{1}{(x_{k+1} - x_0)} \left[\frac{f(x_{k+1})}{(x_{k+1} - x_1) \dots (x_{k+1} - x_k)} - \frac{f(x_0)}{(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_k)} \right. \\ &\quad \left. + \sum_{r=1}^k \frac{[x_r - x_0 - (x_r - x_{k+1})]f(x_r)}{(x_r - x_0) \dots (x_r - x_{k+1})} \right] \\ &= \frac{f(x_0)}{(x_0 - x_1) \dots (x_0 - x_{k+1})} + \sum_{r=1}^k \frac{f(x_r)}{(x_r - x_0) \dots (x_r - x_{k+1})} \\ &\quad + \frac{f(x_{k+1})}{(x_{k+1} - x_0) \dots (x_{k+1} - x_k)} \\ f(x_0, x_1, \dots, x_{k+1}) &= \sum_{r=0}^{k+1} \frac{f(x_r)}{(x_r - x_0) \dots (x_r - x_{k+1})} \end{aligned}$$

Thus we see that, if the result is true for $n = k$, then it is true for $n = k+1$. Since it is true for $n = 1$, it is true for all integral n .

We see that the right hand side of Eq. (4.8) remains the same even if we interchange any two of the arguments $x_0, x_1, \dots, x_n$.

Thus the divided differences are symmetric functions of their arguments.

If any two of $x_0, x_1, \dots, x_n$ happen to be equal, we define the divided differences as follows:

$$f(x_0, x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0).$$

Similarly,

$$f(x_0, x_0, x_0) = f''(x_0).$$

4.4.1 DIVIDED-DIFFERENCE TABLE

The first, second and higher divided differences are usually arranged in the form of a table like the one given below:

x	$f()$	$f(,)$	$f(, ,)$	$f(, , ,)$
x_0	$f(x_0)$			
		$f(x_0, x_1)$		
x_1	$f(x_1)$		$f(x_0, x_1, x_2)$	
		$f(x_1, x_2)$		$f(x_0, x_1, x_2, x_3)$
x_2	$f(x_2)$		$f(x_1, x_2, x_3)$	
		$f(x_2, x_3)$		$f(x_1, x_2, x_3, x_4)$
x_3	$f(x_3)$		$f(x_2, x_3, x_4)$	
		$f(x_3, x_4)$		
x_4	$f(x_4)$			

Thus if we draw a line starting at $f(x_1)$ and sloping towards the right, it passes through successive divided differences $f(x_1, x_2)$, $f(x_1, x_2, x_3)$ and so on.

EXAMPLE 4.2: Form the divided difference table for the data

x	1	2	4	7	8
f(x)	-9	-41	-189	9	523

SOLUTION: The divided-difference table will be as follows

x	f(x)	f(,)	f(,,)	f(,,,)	f(,,,,)
1	-9				
2	-41	-32			
4	-189	-74	-14		
7	9	66	28	7	
8	523	514	112	14	1

From the above table, we can easily write down any difference we want. For example to write down $f(4,7,8)$ we draw a diagonal from $f(4) = -189$. Thus $f(4,7) = 66$ and $f(4,7,8) = 112$.

4.4.2 NEWTON'S INTERPOLATION POLYNOMIAL

From the definition and symmetry of $f(x_0, x_1, x_2)$ etc., we can write

$$f(x) = f(x_0) + (x - x_0) f(x, x_0)$$

$$f(x, x_0) = f(x_0, x_1) + (x - x_1) f(x, x_0, x_1)$$

$$f(x, x_0, x_1) = f(x_0, x_1, x_2) + (x - x_2) f(x, x_0, x_1, x_2)$$

$$\dots \dots \dots \dots \dots \dots \dots$$

$$\dots \dots \dots \dots \dots \dots \dots$$

$$\dots \dots \dots \dots \dots \dots \dots$$

$$f(x, x_0, x_1, \dots, x_{n-1}) = f(x_0, x_1, \dots, x_n) + (x - x_n) f(x, x_0, x_1, \dots, x_n)$$

Multiply the second equation by $(x - x_0)$, the third by $(x - x_0)(x - x_1)$, the fourth by $(x - x_0)(x - x_1)(x - x_2)$ and so on, and add. We get

$$f(x) = f(x_0) + (x-x_0)f(x_0, x_1) + (x-x_0)(x-x_1)f(x_0, x_1, x_2) + \dots + (x-x_0)(x-x_1)\dots(x-x_{n-1})f(x, x_0, x_1, \dots, x_n) + R_n \quad (4.10)$$

$$\text{where } R_n = (x-x_0)(x-x_1)\dots(x-x_n)f(x, x_0, x_1, \dots, x_n) \quad (4.11)$$

Let us write (4.10) as

$$f(x) = P_n(x) + R_n(x)$$

$$P_n = f(x_0) + (x-x_0)f(x_0, x_1) + \dots + (x-x_0)\dots(x-x_{n-1})f(x, x_0, x_1, \dots, x_n) \quad (4.12)$$

It is clear that $P_n(x)$ is a polynomial of degree at most n , also $R_n(x_r) = 0$ for $r=0, 1, 2, \dots, n$. Hence we see that $P_n(x_r) = f(x_r)$, $r=0, 1, 2, \dots, n$. Therefore $P_n(x)$ is a polynomial of degree at most n that passes through the $n+1$ given points $(x_r, f(x_r))$, $r=0, 1, \dots, n$. We have already seen that such a polynomial is unique. Thus $R_n(x)$ must be identical with the error term found in (4.6)

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} F(x). \quad (4.13)$$

where, as before,

$$F(x) = (x-x_0)(x-x_1)\dots(x-x_n).$$

From (4.11) and (4.13) we also see that

$$f(x, x_0, x_1, \dots, x_n) = \frac{f^{(n+1)}(\xi)}{(n+1)!}$$

The polynomial $P_n(x)$ is called Newton's interpolation polynomial with divided differences. We emphasize the point that the Lagrangian and Newton's polynomials are identical. However calculations in the case of Newton's polynomial are simpler. Also x_r ($r=1, 2, \dots, n$) does not appear in any of the differences which involve $x_0, x_1, \dots$ upto x_{r-1} . Suppose we have found the Newton's

polynomial $P_n(x)$ which passes through $n + 1$ given points and now wish to add a new point $(x_{n+1}, f(x_{n+1}))$ to the data. All we have to do is to calculate

$$(x-x_0)(x-x_1)\dots(x-x_n) f(x_0, x_1, \dots, x_{n+1})$$

and add this to $P_n(x)$:

$$P_{n+1}(x) = P_n(x) + (x - x_0)(x - x_1)\dots(x - x_n)f(x_0, x_1, \dots, x_{n+1}).$$

However, for the Lagrangian polynomial we have to calculate every term afresh and the labour is increased manyfold.

EXAMPLE 4.3: Find a 4th degree polynomial which passes through the 5 points given below.

x	f(x)
1.0	-9
2.0	-41
4.0	-189
7.0	9
8.0	523

SOLUTION: The table of divided differences is as follows:

x	f(x)	f(,)	f(,,)	f(,,)	f(,,)
1.0	-9				
		-32			
2.0	-41		-14		
		-74		7	
4.0	-189		28		1
		66		14	
7.0	9		112		
		514			
8.0	523				

$$\begin{aligned}
 P_4(x) &= f(x_0) + (x-x_0)f(x_0, x_1) + (x-x_0)(x-x_1)f(x_0, x_1, x_2) + \dots \\
 &= -9 + (x-1)(-32) + (x-1)(x-2)(-14) + (x-1)(x-2)(x-4)(7) \\
 &\quad + (x-1)(x-2)(x-4)(x-7)(1) \\
 &= -9 - 32(x-1) - 14(x^2 - 3x + 2) + 7(x^3 - 7x^2 + 14x - 8) \\
 &\quad + (x^4 - 14x^3 + 63x^2 - 106x + 56) \\
 &= -5 + 2x + 0x^2 - 7x^3 + 4x^4
 \end{aligned}$$

polynomial $P_n(x)$ which passes through $n + 1$ given points and now wish to add a new point $(x_{n+1}, f(x_{n+1}))$ to the data. All we have to do is to calculate

$$(x-x_0)(x-x_1)\dots(x-x_n) f(x_0, x_1, \dots, x_{n+1})$$

and add this to $P_n(x)$:

$$P_{n+1}(x) = P_n(x) + (x-x_0)(x-x_1)\dots(x-x_n)f(x_0, x_1, \dots, x_{n+1}).$$

However, for the Lagrangian polynomial we have to calculate every term afresh and the labour is increased manyfold.

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SOLUTION: The table of divided differences is as follows:

x	f(x)	f(,)	f(,,)	f(,,)	f(,,,))
1.0	-9				
		-32			
2.0	-41		-14		
		-74		7	
4.0	-189		28		1
		66		14	
7.0	9		112		
		514			
8.0	523				

$$\begin{aligned}
 P_4(x) &= f(x_0) + (x-x_0)f(x_0, x_1) + (x-x_0)(x-x_1)f(x_0, x_1, x_2) + \dots \\
 &= -9 + (x-1)(-32) + (x-1)(x-2)(-14) + (x-1)(x-2)(x-4)(7) \\
 &\quad + (x-1)(x-2)(x-4)(x-7)(1) \\
 &= -9 - 32(x-1) - 14(x^2 - 3x + 2) + 7(x^3 - 7x^2 + 14x - 8) \\
 &\quad + (x^4 - 14x^3 + 63x^2 - 106x + 56) \\
 &= -5 + 2x + 0x^2 - 7x^3 + x^4.
 \end{aligned}$$

So, $P_4(x) = x^4 - 7x^3 + 2x - 5.$

Now if we add a point (5, -173) to the data and ask for the polynomial which involves the 6 given points, all we have to do is to calculate one line in the divided difference table and add one term to the already calculated $P_4(x)$. The last line in the new table is

$$5 \quad -173 \quad 232 \quad 141 \quad 29 \quad 5 \quad 1$$

Hence the new polynomial will be

$$\begin{aligned} P_5(x) &= P_4(x) + (x-1)(x-2)(x-4)(x-7)(x-8)(1) \\ &= x^4 - 7x^3 + 2x - 5 + (x^5 - 22x^4 + 175x^3 - 610x^2 + 904x - 448) \\ &= x^5 - 21x^4 + 168x^3 - 610x^2 + 906x - 453. \end{aligned}$$

EXERCISE 4

1. A function $y = f(x)$ is given by the following table.

x	0	2.5069	5.0154	7.52270
f(x)	0.3989423	0.3988169	0.3984408	0.3978138

- Construct the divided differences table for the above data.
 - Find $f(3.7608)$ using Newton's interpolation polynomial.
2. The following table gives the function $y = f(x)$. Calculate $f(323.5)$ by Newton's interpolation polynomial.

x	321.0	322.8	324.2	325.0
f(x)	2.50651	2.50893	2.51081	2.51188

Compare your result with the exact value if $f(x) = \log_{10} x$.

3. The relation between steam pressure P and temperature T is given by the following table. Evaluate the pressure at temperature 372.1° .

T	361°	367°	378°	387°	399°
P	154.9	167.0	191.0	212.5	244.2

4. Compute the value of $\log_{10} 323.5$ by Lagrange's formula taking the data of problem 2. Compare your result with those obtained previously.
5. Construct the Lagrange interpolation polynomial for the function $y = \sin \pi x$, choosing the points

$$x_0 = 0, x_1 = 1/6, x_2 = 1/2.$$

6. Find the second degree Lagrange interpolating polynomial passing through the three points in the following table.

i	x_i	$f(x_i)$
0	0	-5
1	1	1
2	3	25

7. Use appropriate Newton's interpolating polynomials of degree one, two, and three to approximate $f(2.5)$ if

$$\begin{aligned} f(2.0) &= .5103757 & f(2.6) &= .4813306 \\ f(2.2) &= .5207843 & f(2.8) &= .4359160 \\ f(2.4) &= .5104147 \end{aligned}$$

8. Use the Newton's interpolating polynomial of degree 3 or less to approximate $\cos(.750)$ using the values given below.

$$\begin{aligned} \cos(.698) &= .7661 & \cos(.768) &= .7193 \\ \cos(.733) &= .7432 & \cos(.803) &= .6946. \end{aligned}$$